

Development Roadmap

Development Philosophy

Build one feature at a time.

Every phase must follow:

Backend → Testing → Frontend → Integration → Git Commit

Complete Phases

Phase	Goal
0	Planning & Architecture
1	Environment Setup
2	Backend Foundation
3	Basic AI Chat
4	Frontend Chat UI
5	Streaming Responses
6	Chat Persistence
7	Authentication
8	Conversation Management
9	Markdown Rendering
10	File Upload
11	RAG
12	Tool Calling
13	AI Memory
14	Advanced Frontend UX
15	Testing
16	Dockerization

Phase	Goal
17	Deployment
18	Production Optimization

Daily Development Workflow

Step 1

Learn concept.

Step 2

Build backend APIs.

Step 3

Test using Postman.

Step 4

Build frontend UI.

Step 5

Integrate frontend/backend.

Step 6

Debug issues.

Step 7

Commit to GitHub.

Git Commit Examples

```
git commit -m "Implemented AI chat endpoint"
```

```
git commit -m "Added streaming responses"
```

```
git commit -m "Implemented JWT authentication"
```

Final Goal

Build a production-style AI assistant platform suitable for:

- placements
- internships
- portfolio
- GitHub showcase